

ASAN MEMORIAL COLLEGE OF ARTS AND SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

A PROJECT REPORT ON

AI STUDY BUDDY

An Intelligent Personalized Educational Assistant

Submitted in partial fulfillment for the award of Degree of
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

SUBMITTED BY:

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COURSE & INSTITUTION:

Course: B.Sc. Computer Science

Institution: Asan Memorial College of Arts and Science

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1. Abstract

The **AI Study Buddy** is an interactive educational web application designed to enhance modern learning paradigms using Artificial Intelligence. With the increasing volume of academic resources, students often struggle with effective time management, information overload, and instant query resolution.

This project aims to bridge these gaps by providing an intelligent assistant capable of delivering precise answers, generating concise summaries from long document materials, creating automated practice quizzes, and structuring adaptive study schedules. By integrating Natural Language Processing (NLP) techniques, modern machine learning models, and intuitive user interface design, AI Study Buddy creates a customized and effective learning environment tailored to each student's pace.

2. Introduction & Problem Statement

2.1 Background

Traditional study methods rely heavily on manual summarization, static textbooks, and linear timetables. Students frequently face challenges in filtering crucial concepts from lengthy lecture notes or academic papers, leading to inefficient revision sessions.

2.2 Problem Statement

Existing digital educational tools are often fragmented, offering separate functionality for flashcards, notes, and AI query resolution. There is a lack of a unified, student-friendly platform that combines real-time natural language query handling, automatic revision note generation, and self-assessment tools in one coherent interface.

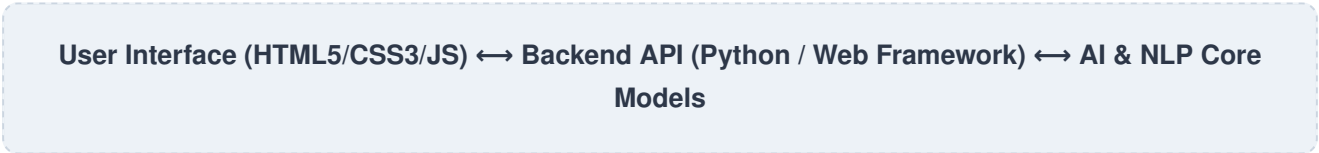
3. Objectives

- **Instant Clarification:** Provide rapid, context-aware answers to student academic queries.
- **Automated Summarization:** Convert lengthy PDF documents, articles, and lecture notes into key takeaways and bullet points.
- **Self-Assessment:** Automatically generate dynamic multiple-choice quizzes and flashcards based on input study topics.
- **User-Centric Experience:** Offer a seamless and responsive web application accessible across both mobile and desktop platforms.

Project Vision: To empower students by leveraging AI automation, making self-paced learning more efficient, tailored, and engaging.

4. System Architecture & Tech Stack

The platform follows a modern client-server architectural model. The front-end interface communicates securely with the backend API service, which processes natural language queries via machine learning algorithms and returns formatted responses.



4.1 Technology Stack

Component	Technology / Framework Used	Purpose
Frontend	HTML5, CSS3, JavaScript	Provides an intuitive and responsive visual UI.
Backend Framework	Python (Flask / FastAPI)	Handles API routes, business logic, and request routing.
AI / NLP Engine	Python ML Libraries / LLM APIs	Processes language queries, summarization, and quiz generation.
Data Management	SQLite / JSON Store	Stores user sessions, saved notes, and quiz results.

5. Core Features & Capabilities

- **Interactive Q&A Bot:** Answers concept-level questions with step-by-step logic and clear explanations.
- **Smart Text Summarizer:** Extracts essential points from long texts, reducing review time significantly.
- **Dynamic Quiz Generator:** Constructs custom quizzes with multiple-choice options and explanatory answers.
- **Study Planner:** Structures topic breakdown lists based on upcoming exam dates and target subjects.

6. Implementation Methodology

The development process was conducted through structured software engineering phases:

- **Requirements Analysis:** Identified core feature needs, student challenges, and interface requirements.
- **Design & Prototyping:** Drafted UI wireframes and structured backend API endpoints.
- **Module Integration:** Integrated Python AI processing backend with the interactive front-end web layout.
- **Testing & Refinement:** Evaluated response accuracy, interface responsiveness, and system response latency.

7. Future Scope & Enhancements

The **AI Study Buddy** project holds significant potential for future expansion:

- **Voice Assistant Support:** Integration of Speech-to-Text and Text-to-Speech capabilities for hands-free interactive studying.
- **Multi-Language Support:** Localizing response engines to support regional languages like Tamil.
- **Collaborative Group Study:** Enabling real-time room sharing for peer discussions and collaborative AI quizzes.

8. Conclusion

The **AI Study Buddy** project successfully demonstrates how Artificial Intelligence can be integrated into academic workflows to improve student productivity. By consolidating information processing, query answering, and dynamic testing into a unified application, the platform streamlines self-directed learning for computer science students and beyond.

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